GadgetZone - Full-Featured E-Commerce Website

Build a fully functional, modern e-commerce platform for **gadgets** using PHP and MySQL. The system should include essential features such as product listings, user authentication, shopping cart, order management, and a responsive UI. Use high-quality product images sourced from **https://unsplash.com**

📁 File Structure

```
gadgetzone/
├── index.php                    # Home page
├── database_setup.sql           # Database schema & sample data
├── migration_stripe_currency.sql  # Migration for settings + Stripe columns
├── includes/
│   ├── db.php                   # Database connection (starts session)
│   ├── functions.php            # Helper functions (auth, cart, formatting)
│   ├── currency.php             # Multi-currency config & formatPrice()
│   ├── header.php               # Shared header (navbar, search, cart badge)
│   └── footer.php               # Shared footer (cache-busted main.js)
├── pages/
│   ├── shop.php                 # Product listing with filters & pagination
│   ├── product.php              # Product detail page
│   ├── cart.php                 # Shopping cart (server-side remove fallback)
│   ├── cart_action.php          # AJAX cart handler (add/update/remove)
│   ├── checkout.php             # Checkout (COD, bKash, Nagad, Stripe)
│   ├── stripe_checkout.php      # Creates Stripe Checkout session
│   ├── stripe_return.php        # Handles Stripe payment callback
│   ├── order_success.php        # Order confirmation page
│   ├── myaccount.php            # User account dashboard (orders, profile, avatar)
│   ├── login.php                # Login page
│   ├── register.php             # Registration page
│   └── logout.php               # Logout handler
├── admin/
│   ├── index.php                # Admin dashboard (stats, recent orders)
│   ├── products.php             # Product CRUD management
│   ├── orders.php               # Order management & status updates
│   ├── users.php                # User & role management
│   ├── settings.php             # Currency & Stripe key configuration
│   ├── layout.php               # Shared admin sidebar (logo links to storefront)
│   ├── footer.php               # Closing tags + admin.js
```

```
│   ├── admin.css          # Admin-only stylesheet
│   ├── admin.js           # Modal, sidebar, image preview JS
│   └── uploads/           # Product image uploads
└── assets/
    ├── css/
    │   └── style.css      # Main stylesheet
    ├── js/
    │   └── main.js        # JavaScript (auto-detects base path for AJAX)
    └── uploads/
        └── avatars/       # User profile picture uploads
```

🏠 HOME PAGE PROMPT

DESIGN AESTHETIC:
- Dark theme background (#0a0a0f) with neon amber accents (#f59e0b)
- Typography: IBM Plex Sans font for headings (bold, 800 weight), DM Sans for body text
- Modern, editorial-style layout with generous spacing
- Smooth animations and hover effects

SECTIONS TO INCLUDE:

1. HERO SECTION:
- Split layout: Left side with headline, description, CTA buttons, stats
- Headline: "Your World. Next-Level Technology."
- Two CTAs: "Shop Now" (primary amber button) and "Explore Deals" (outline button)
- Stats row: "500+ Products", "50K+ Happy Customers", "4.9★ Average Rating"
- Right side: Large product image with floating badge showing "Hot Deal Today - Up to 40% Off"

2. FEATURE STRIP:
- Horizontal bar with 5 features: Free Delivery, 7-Day Returns, 2-Year Warranty, 24/7 Support, Secure Payment
- Each feature has an icon, title, and subtitle
- Background: slightly lighter than main background

3. CATEGORY GRID:

- 6 categories in a grid: Smartphones 📱, Laptops 💻, Audio 🎧, Cameras 📷, Wearables ⌚, Accessories 🔌
- Each card shows emoji icon, category name, and item count
- Hover effect: border changes to amber, slight lift animation

4. FEATURED PRODUCTS:
- Grid of 6 product cards
- Each card shows: image, category label, product name, star rating, price (current + old crossed out), "Add to Cart" button
- Product badge in top-left corner (NEW/HOT/SALE)
- Hover: card lifts up with amber glow shadow

5. DEAL OF THE DAY:
- Large banner with 3 columns: product info (left), product image (center), rating/delivery info (right)
- Countdown timer showing hours:minutes:seconds
- Large price display
- "Add to Cart" and "View Shop" buttons

6. NEW ARRIVALS:
- Similar product grid as Featured Products, showing 4 newest products

7. TESTIMONIALS:
- 3 customer testimonial cards in a row
- Each shows: 5-star rating, review text, customer avatar (circular with initials), name and location

8. NEWSLETTER:
- Full-width amber background section
- Centered content: headline "Get Exclusive Deals First 🎉", description, email input + subscribe button
- Input and button inline (rounded pill shape)

TECHNICAL REQUIREMENTS:
- PHP backend fetching products from MySQL database
- Products table structure: id, category_id, name, slug, description, price, old_price, image_url, badge (NEW/HOT/SALE), stock, featured, created_at
- Categories table: id, name, slug, icon
- Users table: id, first_name, last_name, email, password, phone, address, city, avatar, role, created_at

- `avatar VARCHAR(255)` stores filename of uploaded profile picture (stored in assets/uploads/avatars/)
- `role` ENUM: 'member', 'admin', 'super_admin'
- Settings table: id, setting_key, setting_value (stores active_currency, stripe_public_key, stripe_secret_key)
- Session-based cart stored in $_SESSION['cart'] as [product_id => quantity]

CART IMPLEMENTATION (CRITICAL):
- Cart remove button MUST be wrapped in a <form method="POST"> — this is the server-side fallback
- cart.php handles POST with remove_id at the top BEFORE any HTML output, then redirects
- The remove form is ALSO intercepted by JS (form submit event, not button click) with AJAX for smooth UX
- If AJAX fails for any reason, the form naturally submits and removes the item server-side
- cart badge is ALWAYS in the DOM (not conditionally rendered), hidden via style="display:none" when count=0 — this allows JS to update it without a page reload

JAVASCRIPT BASE PATH (CRITICAL):
- main.js auto-detects environment at the top:
  `const _BASE = window.location.pathname.startsWith('/gadget') ? '/gadget' : '';`
- All AJAX calls use: fetch(_BASE + '/pages/cart_action.php', ...)
- This makes the same JS file work on localhost (/gadget/...) AND production VPS (root /)
- footer.php loads main.js with cache-busting: `<script src="/gadget/assets/js/main.js?v=<?= filemtime(...) ?>"></script>`

ADMIN ACCESS:
- Admin layout.php checks role IN ('admin','super_admin'), redirects to login if not authorized
- Admin sidebar logo links to /gadget/index.php (storefront homepage)
- Default super_admin: email=admin@gadgetzone.com, password=Admin@1234 (bcrypt hashed in DB)

DEPLOYMENT (LOCALHOST vs VPS):
- All PHP paths use /gadget/ prefix for localhost XAMPP (project in htdocs/gadget/)
- On VPS (deployed to domain root), run: find . \(-name "*.php" \) -exec sed -i 's|/gadget/|/|g' {} \;

- Also run the sed command on main.js OR rely on the _BASE auto-detection (recommended)
- Responsive design for mobile/tablet
- Animations: fade-in on scroll for product cards, smooth hover transitions
- Product cards link to product detail page (pages/product.php?slug=...)
```

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### 🛍️ SHOP PAGE PROMPT

```
Create a complete shop/catalog page for an e-commerce website with filtering, sorting, and pagination:

**LAYOUT:**
- Two-column layout: Sidebar (260px) + Main content area
- Breadcrumb navigation at top
- Results count and sort dropdown in main header

**SIDEBAR FILTERS:**
- Sticky sidebar that stays visible on scroll
- Category filter: Radio buttons for All, Smartphones, Laptops, Audio, Cameras, Wearables, Accessories
- Price range slider: Min $0 to Max $300,000 with live label update
- "Apply Filters" and "Clear All" buttons at bottom
- Dark surface background with subtle border

**MAIN CONTENT:**
- Header showing: "Showing 1-9 of 45 results in Smartphones"
- Sort dropdown: Newest, Most Popular, Top Rated, Price Low-High, Price High-Low
- Product grid (3 columns on desktop, responsive to 1 column on mobile)
- Same product card design as home page
- Empty state: Large search icon, "No products found" message, "Clear Filters" button

**PAGINATION:**
- Centered pagination controls at bottom
- Previous/Next arrow buttons
- Page number buttons (current page highlighted in amber)

TECHNICAL FEATURES:
- URL parameters:
?cat=smartphones&search=iphone&sort=price_asc&badge=SALE&page=2
- SQL query with WHERE filters, ORDER BY, LIMIT/OFFSET for pagination
- Form auto-submit on filter change
- Preserve filters (including badge) when sorting/paginating
- Product count per category shown in sidebar
- Sort options: Newest, Most Popular, Top Rated, Price Low-High, Price High-Low
- badge GET param filters by product badge (NEW/HOT/SALE) — used by "Explore Deals" button
- Product card images and names are clickable links to pages/product.php?slug=...

PHP LOGIC:
- Build dynamic WHERE clause from GET parameters
- Calculate total pages: ceil(totalRecords / perPage)
- Fetch products with JOIN to categories table
- Display results or empty state based on query results
```

🛒 CART PAGE PROMPT

```
Create a shopping cart page with dynamic quantity controls and order summary:

**LAYOUT:**
- Two-column: Cart items table (left) + Order summary sidebar (right, 360px)
- Breadcrumb: Home › Shopping Cart
- Page title with item count

**CART ITEMS TABLE:**
- Table header: Product | Price | Quantity | Subtotal | [Remove]
- Each row shows:
  - Product thumbnail (72x72px, rounded corners)
  - Product name and category
  - Unit price (bold, amber color)
  - Quantity controls: [−] [input field] [+] buttons
  - Subtotal (price × quantity)
  - Remove button (× icon)

- Hover effect: slight background highlight
- "Continue Shopping" button below table

**QUANTITY CONTROLS:**
- Minus button, number input (centered), plus button
- Min: 1, Max: 99
- AJAX update on change (no page reload)
- Updates subtotal and order total in real-time

**ORDER SUMMARY (STICKY SIDEBAR):**
- Section title: "Order Summary"
- Subtotal row
- Shipping row: "Free" if order > $5,000, else $150
- Progress message: "Add $X more for free shipping!" (if under threshold)
- Total row (larger, bold, amber color)
- Coupon code input + Apply button
- "Proceed to Checkout" button (full width, large, primary amber)
- Payment icons at bottom: Visa, Mastercard, PayPal, Payoneer
- Security badge: 🔒 Secure Checkout Guaranteed

**EMPTY CART STATE:**
- Centered content: Shopping bag icon (large), "Your cart is empty" heading, description, "Start Shopping" button

AJAX FUNCTIONALITY:
- Add to cart from shop/home pages
- Update quantity
- Remove item (with fade-out animation)
- All cart operations update cart badge in header

PHP/SESSION:
- Cart stored in $_SESSION['cart'] as array [product_id => quantity]
- Functions: getCart(), addToCart(), updateCartQty(), removeFromCart(), getCartCount(), getCartTotal()
- JSON responses for AJAX: {success: true, cart_count: 5, formatted_total: "৳59,999"}
```

💳 CHECKOUT PAGE PROMPT
```

```
Create a multi-step checkout page with order review and payment options:

## LAYOUT:
- Two-column: Checkout form (left) + Order review sidebar (right, sticky)
- Breadcrumb: Home › Cart › Checkout
- Three numbered sections in left column

## SECTION 1: CONTACT INFORMATION
- Step number badge: "1" (amber circle)
- Section title: "Contact Information"
- Form fields in 2-column grid:
  - First Name * | Last Name *
  - Email Address * | Phone Number *
- Asterisk (*) indicates required

## SECTION 2: SHIPPING ADDRESS
- Step number: "2"
- Title: "Shipping Address"
- Fields:
  - Street Address * (full width)
  - City * | Country (Bangladesh, readonly)
  - Order Notes (optional, textarea, 3 rows)

## SECTION 3: PAYMENT METHOD
- Step number: "3"
- Title: "Payment Method"
- Radio button options (custom styled cards):
  - Cash on Delivery 💵
  - bKash (with bKash badge)
  - Nagad (with Nagad badge)
  - Credit/Debit Card 💳
- Each option is a card with border, hover changes border to amber

## ORDER REVIEW SIDEBAR (STICKY):
- Step badge: "✓"
- Title: "Order Review"
- List of cart items:
  - Each item: small thumbnail (56x56), product name, quantity, subtotal
```

- Summary rows:
  - Subtotal
  - Shipping (Free or ৳150)
  - Total (large, bold, amber)
- "Place Order" button showing final total: "Place Order – ৳59,999"
- Security message: 🔒 Your information is secure & encrypted

FORM VALIDATION:
- Server-side validation in PHP
- Error messages displayed at top in alert box
- Required field checks
- Email format validation
- Phone number format

ORDER PROCESSING:
- Generate unique order number: "GZ-" + uniqid()
- Insert into orders table: user_id, order_number, total_amount, status ('pending'), payment_method, shipping_address
- Insert order items into order_items table
- Clear cart session
- Redirect to order success page

PRE-FILL LOGIC:
- If user is logged in, pre-fill form with user data from database
- If not logged in, allow guest checkout
```

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### 👤 MY ACCOUNT PAGE PROMPT

```
Create a user account dashboard with sidebar navigation and multiple tabs:

**LAYOUT:**
- Two-column: Account sidebar (260px, sticky) + Content area
- Breadcrumb: Home › My Account

**SIDEBAR:**
- Top section (card with gradient background):

  - Avatar circle with user initials (2 letters, amber background)
  - User full name
  - Email address
- Navigation menu (vertical list):
  - 📊 Dashboard
  - 🛍️ My Orders
  - 👤 Profile
  - 🔒 Change Password
  - 🚪 Logout (red color)
- Active tab highlighted with amber background

**TAB 1: DASHBOARD**
- Welcome message: "Welcome back, [FirstName]! 👋"
- 3 stat cards in a row:
  - Total Orders (count)
  - Delivered (count)
  - Total Spent (formatted price)
- Recent Orders section:
  - Table showing latest 5 orders
  - Columns: Order #, Date, Total, Status (badge), Payment
  - Status badges color-coded: pending (amber), processing (blue), shipped (purple), delivered (green), cancelled (red)

**TAB 2: MY ORDERS**
- Full orders table (all orders, not just 5)
- Includes "Items" column showing count
- Full timestamp with time
- Scrollable if many orders

**TAB 3: PROFILE**
- Form to update profile information:
  - First Name | Last Name (2 columns)
  - Email (disabled, with note "Email cannot be changed")
  - Phone Number
  - Address (textarea)
  - City
- "Save Changes" button
- Form pre-filled with current user data
- POST to same page, updates database, shows success message

**TAB 4: CHANGE PASSWORD**
- Form fields:
  - Current Password *
  - New Password * (min 6 characters)
  - Confirm New Password *
- Validation:
  - Check current password matches database hash
  - Check new password length
  - Check confirmation matches
- "Update Password" button
- Success/error messages

PHP LOGIC:
- requireLogin() function redirects to login if not authenticated
- getCurrentUser() fetches user from database based on session
- Handle form submissions with POST
- Update user record in database
- Password hashing with password_hash() and verification with password_verify()
- Fetch orders with JOIN to get order items count
```

🔐 LOGIN & REGISTER PROMPTS

LOGIN PAGE:
```
Create a centered login form with:
- Card container (460px max width, dark surface, rounded corners, centered on page)
- Title: "Welcome Back 👋"
- Subtitle: "Log in to your GadgetZone account"
- Form fields:
  - Email Address (email input)
  - Password (password input)
- "Log In" button (full width, large, amber primary button with lock icon)
- Link at bottom: "Don't have an account? Create one →"
- Error messages displayed in red alert box if login fails
- Validation: check email exists, verify password hash
- On success: set $_SESSION['user_id'] and redirect to My Account or original page
```
```

REGISTER PAGE:
```
Create a registration form with:
- Similar card design as login
- Title: "Create Account 🚀"
- Subtitle: "Join GadgetZone and start shopping"
- Form fields:
  - First Name | Last Name (2 columns)
  - Email Address
  - Password (min 6 characters)
  - Confirm Password
- "Create Account" button
- Link: "Already have an account? Log in →"
- Validation:
  - Check all fields filled
  - Email format valid
  - Password minimum 6 characters
  - Passwords match
  - Email not already registered (query database)
- On success: insert user into database with password_hash(), auto-login, redirect to My Account
```

🎨 CSS Custom Properties

The design uses these CSS variables (defined in `:root`):

```css
--accent: #f59e0b;          /* Main amber color */
--accent-light: #fcd34d;    /* Lighter amber for hovers */
--bg: #0a0a0f;              /* Main dark background */
--surface: #16161f;         /* Card backgrounds */
--border: rgba(255,255,255,0.08); /* Subtle borders */
--text: #f0f0f5;            /* Primary text */
--text2: #9090a8;           /* Secondary text */
--font-head: 'IBM Plex Sans';       /* Headings font */
--font-body: 'DM Sans';     /* Body text font */
```

```
--radius: 12px;             /* Border radius */
```

🛠️ Key PHP Functions

In `includes/functions.php`:

```php
isLoggedIn()             // Check if user is logged in
requireLogin()           // Redirect to login if not authenticated
getCurrentUser()         // Get current user data from database
getCart()                // Get cart array from session
addToCart($id, $qty)     // Add product to cart
updateCartQty($id, $qty) // Update cart item quantity
removeFromCart($id)      // Remove item from cart
getCartCount()           // Total items in cart
getCartTotal()           // Total price of cart
formatPrice($price)      // Format as ৳ X,XXX
sanitize($data)          // Clean input data
generateOrderNumber()    // Create unique order ID
```

📱 Responsive Breakpoints

- **Desktop**: > 1024px (full layout)
- **Tablet**: 768px - 1024px (sidebars stack, 2-column grids)
- **Mobile**: < 768px (single column, hamburger menu, simplified tables)

🔒 Security Features

- Password hashing with `password_hash()` and `PASSWORD_DEFAULT`
- SQL injection prevention with `real_escape_string()` and prepared statements
- XSS prevention with `htmlspecialchars()` on all user input
- Session-based authentication
- CSRF protection (can be enhanced with tokens)

🚧 Future Enhancements

- Product search autocomplete
- Wishlist functionality

- Email notifications for orders
- Product reviews and ratings system
- Advanced filters (brand, price range slider)
- Image gallery for products

💳 Stripe Payment Integration

Database Migration

If you are setting up a fresh install, `database_setup.sql` already includes everything. For an existing database, run the migration:

```sql
USE gadgetzone;

-- Settings table (stores currency & Stripe keys)
CREATE TABLE IF NOT EXISTS settings (
    id           INT AUTO_INCREMENT PRIMARY KEY,
    setting_key   VARCHAR(100) NOT NULL UNIQUE,
    setting_value TEXT         NOT NULL,
    updated_at    TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
);

-- Stripe columns on orders table
ALTER TABLE orders
    ADD COLUMN IF NOT EXISTS stripe_session_id VARCHAR(200) DEFAULT NULL,
    ADD COLUMN IF NOT EXISTS payment_status     ENUM('unpaid','paid','refunded')
DEFAULT 'unpaid';

-- Default settings
INSERT INTO settings (setting_key, setting_value) VALUES
    ('active_currency',       'BDT'),
    ('stripe_publishable_key','pk_test_REPLACE_WITH_YOUR_KEY'),
    ('stripe_secret_key',     'sk_test_REPLACE_WITH_YOUR_KEY'),
    ('stripe_webhook_secret', '')
ON DUPLICATE KEY UPDATE setting_value = setting_value;
```

Run via phpMyAdmin: Select `gadgetzone` DB → SQL tab → paste → Go.

Run via CLI:
```bash
mysql -u root -p gadgetzone < migration_stripe_currency.sql
```

Getting Stripe Sandbox Keys

1. Go to dashboard.stripe.com and sign in
2. Enable **Test Mode** (toggle top-right)
3. Go to **Developers → API keys**
4. Copy your keys:
   - **Publishable key** — starts with `pk_test_`
   - **Secret key** — starts with `sk_test_`

Configuring Keys in Admin

1. Login at `/gadget/admin/index.php` (`admin@gadgetzone.com` / `Admin@1234`)
2. Click **Settings** in the sidebar
3. Paste your `pk_test_…` and `sk_test_…` keys in the Stripe section
4. Click **Save All Settings**

Stripe Test Cards

Card	Number	Exp	CVC
Visa (success)	`4242 4242 4242 4242`	`12/25`	`123`
Mastercard	`5555 5555 5555 4444`	`12/25`	`123`
Amex	`3782 822463 10005`	`12/25`	`1234`
Declined	`4000 0000 0000 0002`	`12/25`	`123`

Payment Flow

1. Customer selects **Credit / Debit Card** at checkout
2. Form submits to `pages/stripe_checkout.php` → creates a Stripe session
3. Customer is redirected to Stripe's hosted payment page
4. On success → `pages/stripe_return.php` → verifies session → clears cart → `order_success.php`

5. On cancel → redirected back to checkout with error message

Stripe Files

File	Purpose
`pages/stripe_checkout.php`	Creates Stripe Checkout session
`pages/stripe_return.php`	Handles post-payment callback
`pages/checkout.php`	Payment form (shows Stripe option when keys are configured)
`admin/settings.php`	UI for saving Stripe keys
`migration_stripe_currency.sql`	DB schema for settings + Stripe columns

Troubleshooting

- **Stripe not showing at checkout** — Keys not saved or don't start with `pk_test_` / `sk_test_`
- **"Invalid currency"** — Go to Admin → Settings → select a supported currency
- **"Payment not completed"** — Session expired (valid 24 hrs); retry or use another method
- **Connection issues** — Ensure `curl` is enabled in PHP (`phpinfo()`); check firewall isn't blocking `api.stripe.com`

🌐 Multi-Currency Support

All prices are **stored in BDT (Bangladeshi Taka)** in the database and converted on the fly for display and Stripe.

Supported Currencies

Code	Symbol	Name	Rate (approx)
BDT	৳	Bangladeshi Taka	1.00 (base)
USD	$	US Dollar	×0.0091
EUR	€	Euro	×0.0084
GBP	£	British Pound	×0.0072
CAD	C$	Canadian Dollar	×0.0124
AUD	A$	Australian Dollar	×0.0140

INR	₹	Indian Rupee	×0.76
SGD	S$	Singapore Dollar	×0.0122
SAR	ريال	Saudi Riyal	×0.034
AED	د.إ	UAE Dirham	×0.033
JPY	¥	Japanese Yen	×1.39
MYR	RM	Malaysian Ringgit	×0.042

Changing the Active Currency

1. Admin Dashboard → **Settings**
2. Select a currency from the visual grid
3. Click **Save All Settings**

The change takes effect site-wide immediately (session cache is invalidated on save).

Key Functions (`includes/currency.php`)

```php
getActiveCurrency(): array       // Returns active currency array from DB-backed session cache
formatPrice(float $bdt): string  // Converts BDT → active currency and formats with symbol
convertAmount(float $bdt): float // Converts BDT → active currency numeric value
getStripeAmount(float $bdt): int // Converts BDT to Stripe smallest unit (cents)
getStripeCurrencyCode(): string  // Returns Stripe-compatible currency code (e.g. 'usd')
```

> **Note for Bangladesh merchants:** Stripe does not directly process BDT. When a customer pays by card, the amount is converted to the active display currency (default USD) for Stripe. COD, bKash, and Nagad payments are unaffected and always work in BDT.

📞 Support

For questions or issues, contact: support@gadgetzone.com

📄 License

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